Enhancement Summary — Andre Miranda

CS-499 Computer Science Capstone

Overview of the Artifact

The artifact I selected for enhancement throughout the CS-499 capstone is the Grazioso Salvare Rescue Dashboard, originally developed in CS-340. This full-stack application integrates a Python-based CRUD module, a MongoDB database, a Plotly Dash dashboard, a scoring engine, and a comprehensive testing suite. The original project already demonstrated strengths in secure CRUD operations, data visualization, and rescue-specific filtering logic. However, it also provided significant opportunities for improvement in software engineering, algorithms, and database design.

Throughout the capstone, I completed three major enhancements aligned with the required CS program outcomes:

1. Software Design & Engineering
2. Algorithms & Data Structures
3. Databases

Enhancement One — Software Design & Engineering

The first enhancement focused on transforming the project into a more maintainable, secure, and professionally engineered system. I refactored the CRUD module and dashboard into a cleaner, layered architecture that separates presentation, service, repository, and database responsibilities. Key improvements included:

- Implementing custom exception hierarchies for clearer error categorization
- Adding environment-based configuration to remove hardcoded credentials
- Introducing structured logging across all operations
- Improving validation logic, connection health checks, and rescue filter error handling
- Enhancing dashboard callbacks, layout structure, and readability

These changes demonstrate strong software engineering practices, including modular design, secure coding, maintainability, and professional documentation.

Enhancement Two — Algorithms & Data Structures

The second enhancement introduced a new analytical capability: the Rescue Suitability Scoring Engine. Instead of returning simple filtered results, the system now computes a weighted score for each animal based on breed compatibility, age suitability, outcome history, and other attributes. Major algorithmic improvements included:

- Designing a composed scoring algorithm with multiple weighted factors
- Implementing breed normalization using compiled regex and an LRU cache
- Using dictionaries, lists, and Pandas DataFrames for efficient data processing
- Adding ranking logic to prioritize top rescue candidates
- Expanding the testing suite to validate scoring accuracy, normalization behavior, and edge cases

This enhancement demonstrates algorithmic thinking, performance optimization, data structure selection, and real-world decision-support logic.

Enhancement Three — Databases

The third enhancement strengthened the artifact's database functionality and analytics. I expanded the MongoDB integration to support more advanced operations and improved data reliability. Key enhancements included:

- Implementing MongoDB aggregation pipelines for breed, outcome, and age analytics
- Adding indexes to improve query performance on frequently accessed fields
- Introducing schema validation rules to enforce data integrity
- Improving numeric coercion and error handling for inconsistent AAC dataset values
- Redesigning the dashboard's analytics section with KPIs, histograms, and grouped bar charts

These improvements demonstrate database optimization, schema design, secure access control, and analytical processing.